

📘 AI/ML Internship – Day 27 Notes & Tasks

🟢 Module 5: Pandas (Reading CSV Files & Data Selection)

📌 Part 1: Why CSV Files are Important?

👉 In real AI/ML projects:

- Data is mostly stored in CSV files
- Datasets are loaded using Pandas

👉 Example:

- Student records
 - Sales data
 - Hospital datasets
 - Weather reports
-

📌 Part 2: Theory Answers (IMPORTANT)

◆ 1. What is a CSV File?

👉 CSV means:

Comma Separated Values

👉 Data is stored in table format using commas.

◆ 2. Why CSV Files are Used?

👉 CSV files:

- Are simple
 - Store structured data
 - Work with Excel & databases
-

◆ 3. What is read_csv()?

👉 read_csv() is a Pandas function used to load CSV files into DataFrame.

◆ 4. What is Data Selection?

👉 Selecting specific rows or columns from dataset.

◆ 5. Difference Between loc and iloc

loc	iloc
Uses labels	Uses index positions
Name-based	Number-based

📌 Part 3: Reading CSV File

◆ Example CSV File

```
Name,Marks,City
Rahul,85,Kochi
Anu,90,Trivandrum
Arjun,78,Kollam
```

◆ Read CSV File

```
import pandas as pd
```

```
df = pd.read_csv("students.csv")
```

```
print(df)
```

📌 Part 4: Display Functions

◆ Head Function

```
print(df.head())
```

👉 Displays first 5 rows.

- ◆ Tail Function

```
print(df.tail())
```

👉 Displays last rows.

- ◆ Shape Function

```
print(df.shape)
```

👉 Output:

```
(rows, columns)
```

📌 Part 5: Selecting Columns

- ◆ Single Column

```
print(df["Name"])
```

- ◆ Multiple Columns

```
print(df[["Name", "Marks"]])
```

📌 Part 6: Selecting Rows

- ◆ Using loc

```
print(df.loc[0])
```

👉 Selects row by label/index.

- ◆ Using iloc

```
print(df.iloc[1])
```

👉 Selects row by position.

📌 Part 7: Selecting Specific Values

```
print(df.loc[0, "Name"])
```

◆ Using iloc

```
print(df.iloc[1, 2])
```

📌 Part 8: Slicing Data

◆ Row Slicing

```
print(df[0:2])
```

◆ Column Slicing

```
print(df.loc[:, ["Name", "City"]])
```

📌 Part 9: Filtering Data

```
print(df[df["Marks"] > 80])
```

👉 Filters students with marks above 80.

📌 Part 10: Real-World Thinking

👉 In AI:

- CSV files contain datasets
- We select useful columns/features
- We filter important data

👉 Example:

- Students with high marks

- Patients with disease
 - Customers with high sales
-

Part 11: Visualization of DataFrame

```
+-----+-----+-----+
| Name  | Marks | City   |
+-----+-----+-----+
| Rahul | 85    | Kochi  |
| Anu   | 90    | Trivandrum |
| Arjun | 78    | Kollam |
+-----+-----+-----+
```

Tasks for Day 27

Task 1: Theory (Write Answers)

1. What is CSV file?
 2. Why CSV files are important?
 3. What is read_csv()?
 4. Difference between loc and iloc
 5. What is data filtering?
-

Task 2: CSV Practice

1. Create CSV file
 2. Read CSV using Pandas
 3. Print dataset
-

Task 3: Column Selection

1. Select single column

2. Select multiple columns
 3. Print specific column values
-

Task 4: Row Selection

1. Select row using loc
 2. Select row using iloc
 3. Slice rows
-

Task 5: Filtering

1. Filter marks > 80
2. Filter city names
3. Combine conditions